Mon 1 June 2026, 10:00 · Wachsbleiche room 50/315 · Examiner: Kai-Uwe Kühnberger. Focus topics agreed: **(Local) Search · CSP · Machine Learning.** Allowed: **bring anything** (print this + the Paper cheatsheet). No slides required.

Flow of the whole exam

1. **Greeting / settling in (1–2 min)** small talk, breathe, don't sprint off.
2. **Paper presentation (10–15 min)** you speak mostly freely — see talk plan.
3. **Discussion of the paper** Kühnberger probes: method, critique, placement.
4. **Open AI discussion + focus topics** broad AI basics + depth in Search · CSP · ML.
5. **Wrap-up / grade** short feedback.

Steering trick: at the *end* of the talk, deliberately leave **hooks open** that lead into *your* focus topics (classical BFS/DFS/A*, neuro-symbolic hybrids, scaling/ML). The discussion then drifts where you're strong.

The 15 min: 3 blocks

Block	Time	Content
1. Present	∼5 min	RQ → setup → learned algorithm → results
2. Evaluate	∼4 min	strengths + weaknesses — *score here*
3. Contextualize	∼4 min	LLM-reasoning · scaling/emergence · **neuro-symbolic** · classical search

Rule of thumb: a few points *cleanly* > listing everything.

Block 1 — Present (∼5 min)

1. **What & why** *"Transformers Struggle to Learn to Search", Saparov et al. 2024.* Topic = **search**, the basis of reasoning/planning. Question: too little data, too few params, **or an architectural limit?**
2. **Testbed** minimal: **graph connectivity** (DAG, start→goal, output next vertex). = **proof search** in logic (vertices = facts, edges = implications) → a lower bound on competence.
3. **Difficulty = lookahead** how far ahead you must look before safely committing to move 1. **Limitless data** → rules out "too little data".
4. **Core = training distribution** Naïve → only small lookaheads → **shortcuts**. Only **Balanced** (lookahead uniform, shortcuts removed) → learns search **near-perfectly**. *Msg 1: whether a transformer learns search depends heavily on the data.*
5. **Real contribution = mechanistic interpretability** activation patching, reconstruct the computation graph *without* assuming the algorithm → finds **exponential path-merging** (each vertex stores its reachable set; each layer **unions** → doubles → search exponential in #layers ≈ transitive closure in log depth).
6. **3 results** (i) bigger graphs harder, **more params don't help**; (ii) **CoT** (DFS/selection-inference) → constant layers but **still fails** on big graphs; (iii) **conclusion:** scaling alone ≠ robust search → need different training (curriculum) / architecture (looped transformers).

60-sec emergency version: "Small transformers learn graph search near-perfectly — *but only* with an artificially **balanced** distribution that removes shortcuts. Mechanistically: **exponential path-merging** (reachable sets double per layer). Bigger graphs break it — and **neither more params nor CoT** fix it. Conclusion: robust search is **not just a scaling problem**."

Block 2 — Critique *(Learnability ≠ Expressivity)*

- **Title overreaches:** shows a **trainability** failure (seed variance, distribution sensitivity, non-convergence), **not** a representational limit. Merrill & Sabharwal: CoT-transformers **can** represent search (∈ P ⊇ connectivity); the paper's own **positive** result (Balanced) shows the same architecture **learns** search → so it's data/optimization, not architecture.
- **Rebuttal ready:** *"but more params don't help (Fig. 7)!"* → still a **trainability** claim *in the tiny regime*; says nothing about representability / frontier scale.
- **Extrapolation gap:** dim 16, ≤60M params — 3–4 orders below GPT-class.
- *The mech-interp method is a genuine, reusable contribution — stay fair, give credit.*

Block 3 — Contextualize (bridges to your topics)

- **Reasoning vs. pattern-matching** (Kambhampati; Bachmann & Nagarajan).
- **Scaling laws & emergence** — Wei vs. Schaeffer ("mirage") → paper = counterexample to scaling optimism.
- ⋆ **Neuro-symbolic AI** (Kühnberger's turf!): classical search (BFS/DFS/A*) is **provably correct & scales**; learned transformers only approximate → argument for **hybrids** (offload search to symbolic solvers).
- **Bridge — (Local) Search:** path-merging = parallel, transitive-closure-style build-up of *all* vertices at once — *not* sequential frontier expansion like BFS/DFS; ∼log(L) layers. CoT-DFS variant ↔ lecture DFS.

Before the exam: say the Block-1 script **3× out loud with a timer**. Deliver Blocks 2 & 3 freely from bullets. Depth via the paper quiz + "Quiz Exam Schwerpunkte".